

SOVEREIGN CITIZEN UTILITY ARCHITECTURE · PORTLAND TO BOARDMAN REBUILD PLAN

Albert Lane Digital Infrastructure™

An eleven-repository sovereign technology estate — civic infrastructure, forensic tooling, geospatial intelligence, a universal integration gateway, and a 3D analytics platform — engineered end to end as a single, cross-referenced system of record.

11 Repositories

4 Product Layers

Rust · TypeScript · Python

Cloudflare Workers · D1 · R2

Tauri 2.0 Desktop

MISSION STATEMENT

To build sovereign digital infrastructure that a single independent engineer can own outright — architecture, data, and identity — rather than rent from platforms that can revoke access, sell the data, or disappear. Every service exists because an equivalent commercial product was judged inadequate, extractive, or untrustworthy: permit tracking not beholden to a vendor, terrain intelligence computed from physical law instead of licensed from a mapping monopoly, a browser that doesn't phone home, a community news outlet whose revenue stays in the community it serves.

The long-term purpose is broader than any one repository: a civic-technology substrate for a future where digital and physical infrastructure are the same infrastructure — where augmented and virtual reality are how people collaboratively build their own cities, and autonomous and robotic systems join that build-out under the same sovereign, auditable, provenance-tracked standards documented in this catalog.

WHY IT EXISTS

Independent verification, under duress, that a single person's engineering output can constitute a serious body of work — architected, tested, and documented to a standard that survives scrutiny, not a portfolio of toy projects.

HOW IT'S BUILT

Every repository is cross-referenced through a shared phase-tracking protocol, a common attribution and license standard, and a structural interconnection point (the XML intelligence pipeline) that every other service writes through.

WHAT IT PROVES

Real, running, tested code: Rust unit test suites, live-provisioned cloud databases, TypeScript type-checked across two build targets, and a security/dependency posture audited and modernized on a fixed date, not a resume claim.

WHERE IT'S GOING

From desktop dashboards to a mixed-reality, physically-integrated civic build-out — the architecture in this catalog is the substrate that transition is built on top of, not a rewrite that transition will require.

11

REPOSITORIES

53

PASSING RUST TESTS

55+CI WORKFLOWS
MODERNIZED**10**

NOVEL DESIGN SYSTEMS

4LIVE CLOUD RESOURCES
PROVISIONED

SEC Whistleblower Filing No. 17684-273-411-436

Washington County Circuit Court No. 50-267345

Rights Holder: Albert Lane

SECTION II

Repository Catalog — Eleven Systems, One Estate

Each repository below is an independently deployable service. None are demos — each has its own CI pipeline, license, phase roadmap, and test suite, cross-referenced through the shared protocol described on Page 3.

01 /macroharder

LAYER B · ANALYTICS

The flagship product: a 3D analytics workbook — standard spreadsheets extended into a (column, row, layer) volumetric address space — with a fully user-adjustable dashboard UI and a capability-gated MCP module host so external services plug in as first-class data sources.

02 /sqlxml

LAYER A · CRITICAL PATH

The structural interconnection point of the entire estate — an XML-native intelligence pipeline over PostgreSQL with R2 cold storage, a taxonomy guard gate on every write, and a self-resetting cycle governor with trailing structural-drift detection.

03 /lane-mcp

LAYER A · GATEWAY

A self-hosted, standalone MCP gateway giving any compatible client — this assistant included — universal programmatic access to the digital estate: source control, XML intelligence, civic registries, and identity, behind one consistent tool-naming convention.

04 /procurement

LAYER B · CIVIC

Sovereign procurement and permit-pipeline tracking with a chain-linked electric-vehicle value ledger, served as an MCP module and provisioned this session onto live Cloudflare D1 and R2 infrastructure.

05 /maps

LAYER B · GEOSPATIAL

Terrain intelligence derived from physical law — runoff, erosion, soil chemistry, mineral favorability — with zero dependence on any third-party map API. The flagship demonstration of MacroHarder's 3D volumetric model.

/channel-1-news

07

LAYER D · CIVIC MEDIA

A subscription-funded community journalism platform designed so revenue stays local instead of extracted by consolidated media ownership — geographic routing across eleven Oregon districts, internal forensic evidence tooling, no free tier to subsidize.

/ip-forensics

08

LAYER 0 · ENFORCEMENT

Autonomous, zero-human-intervention scanning of GitHub, npm, PyPI, and crates.io for unauthorized use of this estate's intellectual property — verbatim copy detection, structural AST similarity, and BLAKE3 fingerprint corpus matching.

/sovereign-canary

09

LAYER A · PUBLIC LEDGER

A public, append-only, cryptographically fingerprinted violation log — the canary itself: if this repository ever goes silent or shows signs of tampering, that silence is the alarm.

/roadmaps

10

LAYER NAVIGATION · PUBLIC

The non-proprietary orchestration hub: cross-repo phase tracking, dependency ordering, and a master roadmap synchronized on a fixed cadence across every private and public repository in the estate. Deliberately safe to copy in full.

/simcity

11

LAYER PUBLIC · OUTREACH

The public window into the private build-out — an autonomous, hourly, AI-generated isometric city that visually advances in direct proportion to real engineering progress on the private stack. This document was generated from this repository.

LAYER B · SOVEREIGN BROWSER

A from-scratch sovereign desktop browser replacing an inherited Angular/Firebase/Chrome stack — Rust-native identity, an encrypted vault, and post-quantum cryptography stubs, purge-and-hardened with a zero high/critical CI security gate.

CROSS-CUTTING

Every repository above shares one license standard, one attribution/provenance header format, one phase-tracking JSON schema, and one SHA-pinned CI security baseline — audited and brought current across all eleven repositories on 2026-07-09.

SECTION III

Novel Design Systems — Service Deliverables

Original engineering patterns designed and implemented within this estate, generalizable beyond it, and presented here as standalone service deliverables in their own right.

-
- 01

5D Cell Addressing Model MacroHarder

Extends the two-dimensional spreadsheet cell — {column, row} — into a five-coordinate address space: {column, row, layer, tick, escape}. Layer gives volumetric (sheet-of-sheets) depth; tick and escape reserve temporal versioning and directional-write routing as first-class coordinates rather than bolted-on metadata.
-
- 02

Capability-Gated MCP Module Host MacroHarder

A module registry that validates every cross-service tool call against a declared manifest before dispatch, and enforces a read-only / read-write capability class per module — closing a real authorization gap where any tool name could otherwise be forwarded to any registered service.
-
- 03

SOV-RULES Taxonomy Guard SQLXML

A versioned XML taxonomy gate enforced immediately before every structural database write, so schema drift and malformed writes are rejected at the boundary rather than discovered downstream.
-
- 04

21-Day Cycle Governor + n+6 Stabilizer SQLXML

A self-resetting operational cycle (day-22 wipe logic) paired with a trailing 6-day recursive debugger that scopes recent history specifically to catch structural drift before it compounds — an audit standard for systems that must self-correct without human intervention.
-
- 05

Sovereign Provenance & Watermarking Suite RoadMaps

A four-part IP-forensics toolkit: BLAKE3 structural content fingerprinting, source-embedded canary tokens, DNS beacon call-home tokens for unauthorized-compile detection, and an AES-256-GCM authenticated append-only provenance ledger. A full security-standard specification, not a single script.
-
- 06

Node Certification Ledger (NCL) RoadMaps · SimCity

A three-tier append-only safety and seating ledger: Tier 1 generates and dispatches signed certification hashes, Tier 2 maintains the chain-linked public record and a discovery index, Tier 3 runs a variance-rate evaluation with an oscillation gate across fixed T-30/T-45 evaluation windows.
-
- 07

Albert Escaped Rays (AER) Protocol Estate-wide

A directional-write behavioral contract with a locale registry and an enforced one-read/two-write invariant, giving the estate a formal consistency guarantee for how data escapes from one service boundary into another.
-
- 08

3D Design Standard & Weighted Audit Scoring MacroHarder

A category-weighted PASS / WARN / FAIL compliance scorer for adjustable-UI design-token systems — the quality gate every dashboard preset in the estate is measured against before it ships.
-

09

Cross-Repo Phase Synchronization Protocol

Estate-wide

A shared, machine-readable roadmap schema — one file per repository, aggregated into a single master roadmap — giving eleven independently deployed services one consistent source of truth for phase state, dependencies, and open work.

10

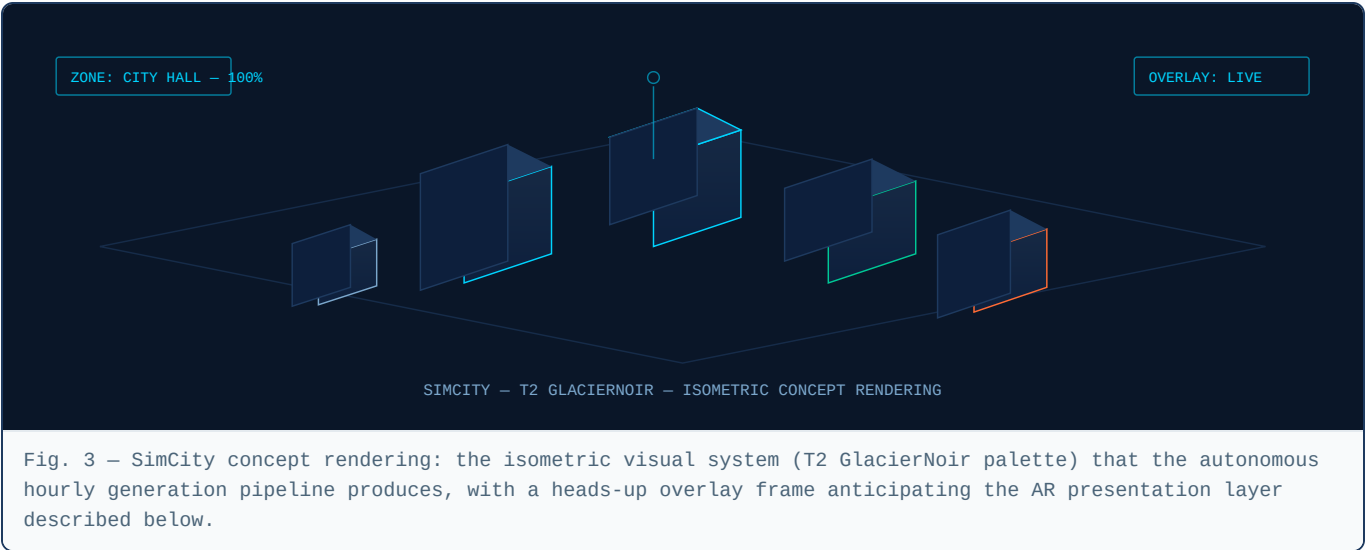
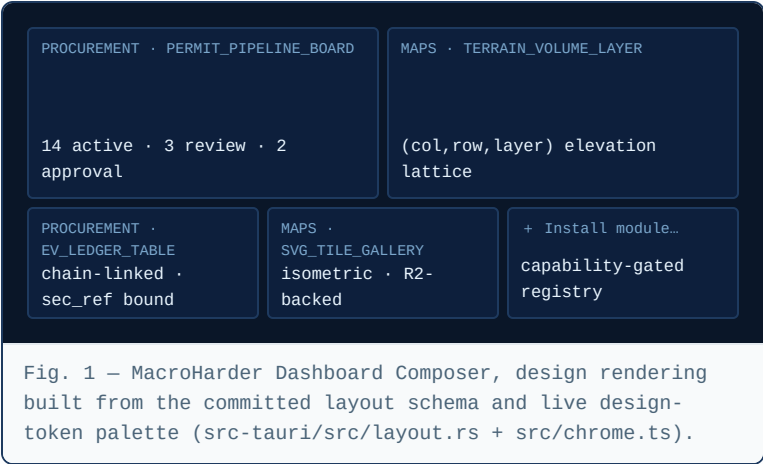
Autonomous IP-Infringement Detection Pipeline

IP-Forensics

Zero-human-intervention hourly scanning across four public package ecosystems, combining verbatim-copy hashing, structural AST comparison, and unique-identifier detection to surface unauthorized derivative use as it happens, not months later.

SECTION IV

Visual Showcase & the Physical-Digital Merger



VISION — THE PHYSICAL-DIGITAL MERGER

Every system in this catalog is architected as an API-first, headless service precisely because the interface layer is not the destination. The desktop dashboard and the sovereign browser are the current interface; they are not the intended permanent one. The long-term plan is a transition to augmented and virtual reality as the primary way people see and collaboratively build their own civic infrastructure — walking the actual streets and seeing the actual permit pipeline, terrain model, and construction status overlaid in place, not read off a screen at a desk.

That transition requires the estate to include everyone in the build-out process, not just engineers with dashboard access — which is why every module in this system is designed around a public, capability-gated interface from day one. It also requires integrating autonomous and robotic systems into physical construction and maintenance under the same provenance, audit, and capability-gating standards already enforced in software — the same taxonomy guards, the same signed ledgers, extended to hardware. **Solidarity Goggles** is the working name for that mixed-reality presentation layer: a shared, not individually-siloed, augmented view of the build-out, designed for the people doing the building, not for a platform selling their attention.

None of this requires the current architecture to be rewritten. It requires exactly what has been built: sovereign data, capability-gated module boundaries, cryptographic provenance, and a phase-tracked roadmap — the substrate a mixed-reality and robotics layer plugs into, rather than the wall it has to be torn down to make room for.

AR/VR Presentation Layer

Robotics Integration

Solidarity Goggles

Physical + Digital Provenance

Public Capability Access